parcha

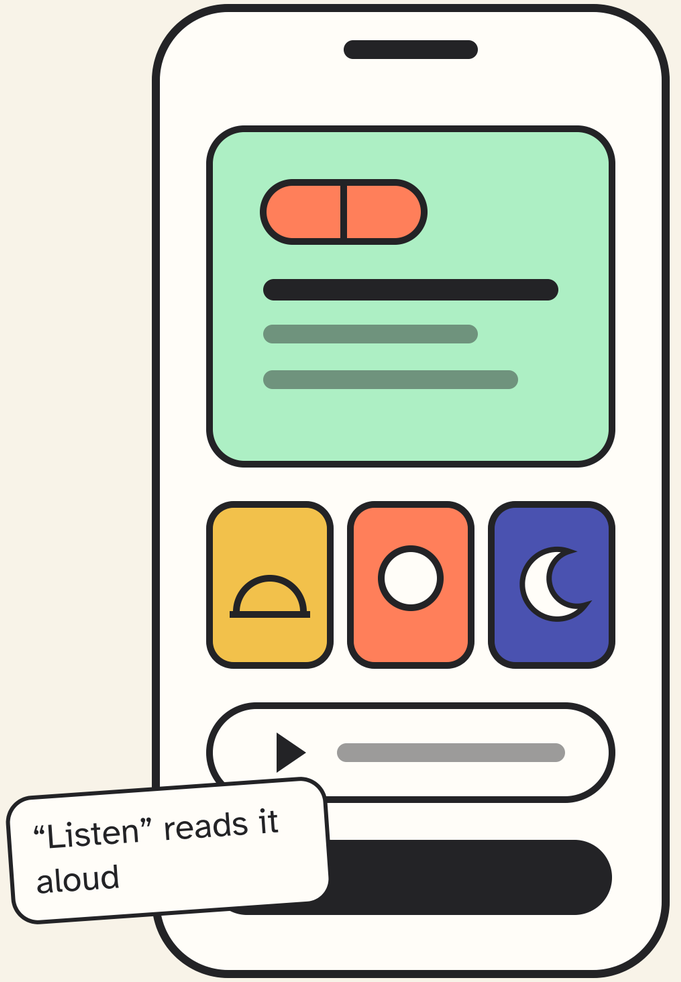
A prescription the patient’s family can read, hear and follow, even offline.

One app for the outpatient visit: the patient asks from their phone, the desk takes vitals, the doctor prescribes with a safety check, and the medicines arrive on the family’s phone as pictures, in Hindi, English or Hinglish.

Clinic · parcha-clinic.web.app

Patient · parcha.web.app

Windows · parcha-desktop.web.app



Where the visit breaks

Outpatient care rarely fails inside the consultation. It fails at the handoffs around it, and small clinics have no IT staff to patch them.

62% of OPD patients waited **over 30 minutes**, and most consultations lasted **5 minutes or less**. *CAG audit, J&K tertiary hospitals*

0 / 75 primary health centres prescribed through an electronic record. Lack of **IT staff, staff and money** were the main barriers. *Study of 75 PHCs*

Net **Unreliable internet** is a top barrier named by doctors in Indian rural hospitals. Software that stops when the line drops gets abandoned. *PMC study, 2025*

THE HANDOFFS, TODAY

1 · **Patient** waits in the room, not knowing when

2 · **Desk** copies the same facts into a register

3 · **Doctor** writes by hand with no safety check

4 · **Family** forgets which medicine, and when

Sources: Daily Excelsior on the CAG audit of J&K OPDs; DOAJ ff817ec8b0a24550b96bb25343ee9c5d (75 PHCs); PMC12087044 (rural hospitals). Studies from India; one region each, not national figures.

One chain, request to medicine taken

1 · Patient Asks a hospital for a visit from the phone and watches a **15-minute clock** on a map.

2 · Desk Finds a doctor who accepts a time. Unanswered in 15 minutes, the request **moves to the next hospital**.

3 · Triage Checks the patient in, takes **vitals and the complaint**, and sends them to a chosen doctor's queue.

4 · Doctor Same as last visit, saved sets, typo-proof search or dictation; tests; a **live safety check**; signs with Ctrl+Enter.

5 · Slip Printed with the doctor's registration number, and on the family's phone as **dose pictures, read aloud**.

6 · Home A day-by-day schedule to tick off, with reminders. Pharmacy and billing are optional seats.

Fits a solo doctor or a hospital

4 set-ups to start from: large hospital, medium hospital, personal practice, or an empty custom clinic.

Roles are made from 7 abilities; one person can hold several, and what they may do adds up.

Management (people, roles, who sends to whom) sits behind a PIN.

Messages inside the team are end-to-end encrypted; patients are shared as links, never copied.

सुबह 1 · रात 1 · खाने के बाद

Safe, private, and offline

7 safety rules	**68** medicines in the list	**53** interaction pairs	**21** cautions on vitals	**37** with age, dose or pregnancy limits

The safety check

Allergy, two medicines together (and last month's), today's vitals, age, the most a day (a child's by weight), pregnancy, and medicines written by hand, flagged as unchecked.

Runs as the doctor writes and **again on the server at signing**. It advises; a high flag must be read before signing.

When the internet drops

Every action **queues on the computer** and is sent when the line returns, never twice.

The safety check runs locally. A slip signed offline prints at once, marked unconfirmed.

The Windows program keeps an **AES-GCM encrypted** copy of the day.

Who sees what

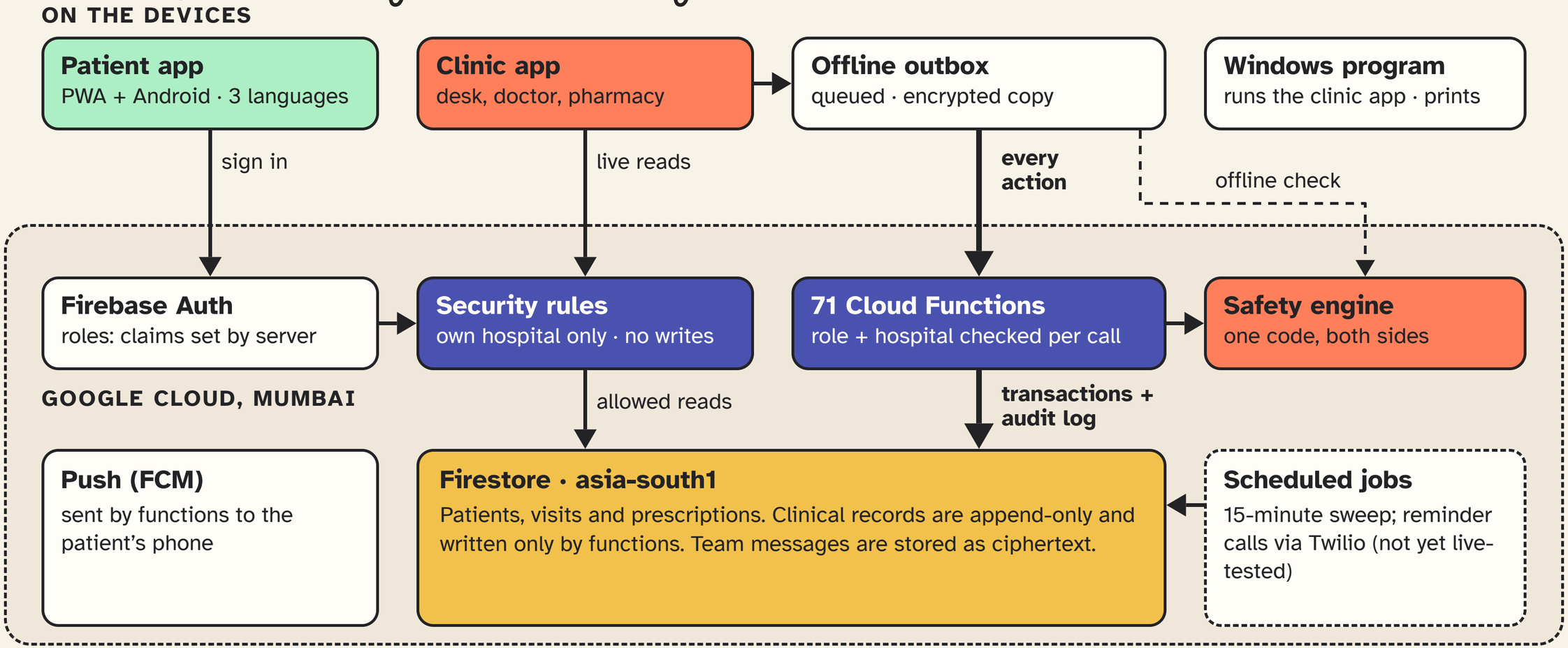
Roles are claims **only the server sets**; clinical records are written only by server code; each hospital sees its own.

Sessions lock after 10 idle minutes; signing asks for the password again.

Data and server code in **Mumbai** (asia-south1).

Counts from the code: packages/core (formulary, special.ts, conflicts.ts). A demo dataset of textbook figures, not a full drug database: a clear result means nothing in this list matched.

Every write goes through the server



Hosted on Firebase: the clinic and patient web apps, the Android download and the Windows installer. One TypeScript core is shared by all three apps and the server.

Tested, honest, and next

Tested

89 unit tests: safety rules, schedules, roles

58 tests of the rules and functions on Firebase emulators

Live with sample data: clinic and patient web apps, an Android download and a Windows installer.

Not proven yet

- No real clinic or patient yet
- Time per prescription not yet measured
- Android app not run on a device
- Reminder calls not run on a live line
- Medicine list is a 68-drug demo set
- Privacy notice not reviewed by a lawyer

Next: one pilot clinic

- Stopwatch: arrival to signed slip
- Doses ticked against doses prescribed
- Reminder calls answered
- Then: WhatsApp sharing, ABHA number, a licensed drug database

Why calls: an automated call plus a picture SMS raised adherence from 85% to 91% in a Bangalore cohort of young urban HIV patients.

Try it live

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Call study: PMC3428344, a different population from older chronic-disease patients, so the pilot measures it again. All live data is sample data.